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Task Description:

The overall goal of the proposed research is to identify the structural and functional changes that occur in the utricle of the vestibular labyrinth as the result of transitions from one gravitational level into another. This information is critical for advancing our understanding of neurovestibular adaptation and its sequelae. Our strategy is to use centrifugation as a tool for temporarily altering the gravitational stimulus. We are particularly interested in determining the afferent firing rate and response characteristics and the morphological changes in the hair cell-afferent complexes found in the utricle that are modified when the organism moves from normal gravity (1G) to hyper-gravity (hyper-G) and in the opposite sense.
In the utricular otolith organ it was found that the afferent response sensitivity to head acceleration (13, fish) as well as the number of synaptic bodies in hair cells (56, rats) increase as the result of exposure to microgravity (μG). A preliminary result showed that exposure to hyper-G by centrifugation decreases the number of synaptic bodies (56, rats). The period of afferent re-adaptation to normal gravity in fish (13) was found to be close to the period of behavioral adaptation in humans (55). Re-adaptation is defined here as the process of adaptation to normal gravity. These results suggest the possibility that the process of adaptation to μG might be mimicked by the process of re-adaptation from a hyper-G level. However, the experiments that would combine the results of μG and hyper-G experiments for the same species are absent. Further, the correlations of the synaptic re-arrangement with afferent sensitivity following exposure to altered gravity for the same species remain undefined. The purpose of the present proposal is to fill these gaps.

Our model is the adult toadfish, Opsanus tau. Fish provide an excellent model system for gravitational research (7). We have everything in place to successfully complete the proposed set of experiments and modeling. Of prime importance is the fact that in the same species a comprehensive evaluation of the neural response to μG (13) and hyper-G (present proposal) exposure will be made. There are four specific aims: Aim 1. Response characteristics of utricular afferents to Centrifugation. Fish will be placed in groups and exposed to different magnitudes (1.5 and 3 g) and duration (1-16 days) of centrifugation. On Center Controls and ground controls are included. Using electrophysiological techniques, individual utricular afferents will be physiologically characterized based on gravitoinertial acceleration testing. Afferent response properties to efferent vestibular activation will be specified. The duration of afferent recording session is tempered by the need to freeze the tissue in time to permit the measurements of structure-function correlations. To match the conditions under which post-flight experiments were performed (13) changes in afferent sensitivity caused by hyper-G will be studied from 1-24 hrs after centrifugation, and re-adaptation after 24-96 hrs in 1G . Aim 2. Morphology of hair cell-afferent synaptic complexes. Combining the techniques of labeling of individual afferents with light and transmission electron microscopy a complete characterization of the afferent is possible: its physiology, its dendritic morphology in the macula, and the extent and quantification of synapses formed from the innervated hair cells. Visualization of complex 3-D microcircuitry will also reveal precise connectivity within the macula that is essential for understanding neurovestibular adaptation. Synaptic body counts will be performed to address the fundamental question of the correlation of afferent response sensitivity with hair cell synapse number. Aim 3. Merging the structure and function of otolith afferents and their relationships to hyper-G. The specific times and thresholds of the afferent response adaptation and re-adaptation and the structural changes associated with the hair cell-afferent synapse will be determined by varying the magnitude and duration of centrifugation. Aim 4. Modeling. Based on known methodologies and the present experimental results, research will be directed towards the establishment of phenomenological models describing the adaptive changes in utricular afferents in altered gravity conditions. We will examine the relationship between the afferent response and the number of synapses in normal and in hyper-G conditions. A model of toadfish otolith mechanics in these conditions will be also developed. The four specific aims are directed to achieve the main goals: 1) to test the correlations of change in the gravitational level and the change in afferent response characteristics caused by head acceleration and the potential underlying changes in the morphology of the associated hair cell-afferent synaptic complexes, 2) to examine the role of efferent vestibular system on these correlations, 3) to test (propose) the possible mechanisms responsible for these correlations; and 4) to test if the transition from hyper-G to normal gravity can be used to mimic the effects of neural adaptation to μG. We expect that convergence of findings in μG and hyper-G will provide a model of neurovestibular plasticity, and will lead to the development of novel test protocols for both ground- and space-based research. As we learn about the basic cellular processes in vestibular neurons, countermeasures will eventually be developed to reduce the symptoms associated with Space Motion Sickness or even modify synaptic sensitivity in preparation of entry into a higher gravity level.

The overarching hypotheses to be evaluated are: One, the gravitoinertial sensitivity of otolith nerve afferents is altered as a consequence of the animal's exposure to hyper-G conditions as well as μG. It is reasoned that the transition from 1G to hyper-G will decrease otolith afferent sensitivity (output in impulses per sec/input in G), and the transition from hyper-G back to 1G will increase it (and back to normal levels). Thus, the transition from hyper-G to normal gravity may serve as an imitator of neurovestibular sequelae to μG. Two, hair cell-afferent synaptic complexes are modified by the altered gravity condition: the up- or down regulation of synaptic bodies is specific to the direction of transition from 1G. The change in number of synapses is directly correlated with afferent response sensitivity. Three, afferent adaptation mechanisms may involve either self-learning processes confined to the hair cell or responses of the hair cell-afferent system to feedback signal(s) that originate from a misbalance of information coming from otolith and extra-otolith contributor(s), or a combination of both. (These mechanisms will be considered by modeling based on the experimental findings.) Four, efferent vestibular actions are not uniform on hair cells and afferents, and it is proposed that the changes brought on in otolith afferents by the transitions from one gravity state to another will be most prevalent and coincident in afferents strongly affected by efferent activation. If this proves to be the case, then the next step is to directly query the efferent neurons themselves. The overall goal of the proposed research is to identify the structural and functional changes that occur in the utricle of the vertebrate vestibular labyrinth as the result of transitions from one gravitational level into another. This information is critical for advancing our understanding of neurovestibular adaptation and its sequelae, and for the effective design and validation of countermeasures to benefit astronaut health and performance. Our strategy is to use centrifugation as a tool for temporarily altering the gravitational stimulus. We are particularly interested in determining the afferent firing rate and response characteristics and the morphological changes in the hair cell-afferent complexes found in the utricle that are modified when the organism moves from normal gravity (1G) to hyper-gravity (hyper-G) and in the opposite sense. This study is a natural analog to our previous studies aboard STS-90 and 95, and will provide a comprehensive evaluation of the neural response to μG and hyper-G in the same species. We expect that convergence of findings in μG and hyper-G will provide a solid model of neurovestibular plasticity, and will lead to the development of novel test protocols for both ground- and space-based research. As we learn about the basic cellular processes in vestibular neurons, countermeasures will eventually be developed to reduce the symptoms associated with Space Motion Sickness, benefit sensorimotor coordination and performance, or even down regulate synaptic sensitivity in preparation of entry into a higher gravity level.

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The functional importance of head stabilization is seen by our ability to change gaze orientation and maintain binocular fixation of objects of interest during not only standing but also during locomotor activities, and our ability to detect self-motion and adaptively react to applied perturbations of the head and body. In the space environment the vestibular system is profoundly affected by the loss of he gravity vector. However, conditions on Earth can seriously affect the individual's ability to functional normally. In addition aging places a serious challenge to the vestibular and neural centers responsible for posture and movement control. Disorders of the vestibular system affect head, gaze and posture stabilization, and can dramatically alter the patients willingness to induce self-motion and can lead to trepidation of normal activities. These studies will also provide important insights into vestibular function, and may help in the diagnosis and treatment of vestibular patients seen in the clinic.
	NASA Grant # NASA 03-OBPR-04 P.I.: Richard Boyle NASA Ames Research Center, SLR Task Title: Structure & Function of Neurovestibular Adaptation & Re-adaptation in the Utricle Following Extended Periods of Hypergravity Exposure Project Start Date: 1 October 2004 – 30 September 2005 Date of Report: 20 May 2005 The Specifics Aims of the project remain as originally presented. We have made one addition to the project that was suggested by the study section panel that we found insightful: to extend the study to determine how the constant velocity rotation experienced during centrifugation influences the discharge and response properties of the horizontal semicircular afferent nerves. In the next section we will present work conducted to meet each of our aims. The overall goal of the proposed research is to identify the structural and functional changes that occur in the utricle of the vertebrate vestibular labyrinth as the result of transitions from one gravitational level into another. This information is critical for advancing our understanding of neurovestibular adaptation and its sequelae, and for the effective design and validation of countermeasures to benefit astronaut health and performance. Our strategy is to use centrifugation as a tool for temporarily altering the gravitational stimulus. We are particularly interested in determining the afferent firing rate and response characteristics and the morphological changes in the hair cell-afferent complexes found in the utricle that are modified when the organism moves from normal gravity (1G) to hyper-gravity (hyper-G) and in the opposite sense. This study is a natural analog to our previous studies aboard STS-90 and 95, and will provide a comprehensive evaluation of the neural response to μG and hyper-G in the same species. 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As we learn about the basic cellular processes in vestibular neurons, countermeasures will eventually be developed to reduce the symptoms associated with Space Motion Sickness, benefit sensorimotor coordination and performance, or even down regulate synaptic sensitivity in preparation of entry into a higher gravity level. The overarching hypotheses to be evaluated are: One, the gravitoinertial sensitivity of otolith nerve afferents is altered as a consequence of the animal's exposure to hyper-G conditions as well as μG. It is reasoned that the transition from 1G to hyper-G will decrease otolith afferent sensitivity (output in impulses per sec/input in G), and the transition from hyper-G back to 1G will increase it (and back to normal levels). Thus, the transition from hyper-G to normal gravity may serve as an imitator of neurovestibular sequelae to μG. Two, hair cell-afferent synaptic complexes are modified by the altered gravity condition: the up- or down regulation of synaptic bodies is specific to the direction of transition from 1G. The change in number of synapses is directly correlated with afferent response sensitivity. Three, afferent adaptation mechanisms may involve either self-learning processes confined to the hair cell or responses of the hair cell-afferent system to feedback signal(s) that originate from a misbalance of information coming from otolith and extra-otolith contributor(s), or a combination of both. (These mechanisms will be considered by modeling based on the experimental findings.) Four, efferent vestibular actions are not uniform on hair cells and afferents, and it is proposed that the changes brought on in otolith afferents by the transitions from one gravity state to another will be most prevalent and coincident in afferents strongly affected by efferent activation. If this proves to be the case, then the next step is to directly query the efferent neurons themselves. Summary of Research Progress and Accomplishments Significant progress has been made on each of the specific aims originally presented and on the new aim to characterize the response of the horizontal semicircular canal afferents to the centrifugation stimulation paradigm. We are seven months into our project; we have completed the majority of the required control experiments; have two (2) abstracts (see next section) at a national and international meeting; and one paper nearly complete to submit for publication (see next section). 1) Control Utricular Afferent Characterization Since the start of the project on October 1 2004, we have recorded the firing rate characteristics of 223 utricular afferents to controlled translational accelerations and dynamic and static tilts from pitch through roll, and their response to activation of the central efferent neurons using electrical pulse stimuli. In 82 afferents a complete characterization was successful; in the remaining 141 afferents the are still relevant and will be used to compile the statistical properties of the discharge characteristics. In these studies the discharge of single utricular afferents is obtained using glass microelectrodes, and acceleration stimuli are applied at systematically varying head (body) orientation to specify the afferent response characteristics. It is imperative to obtain the afferent’s maximum and minimum sensitivity to the applied stimuli for direct comparison of control responses to those obtained following centrifugation. We are currently analyzing and compiling the data obtained to date. We will conduct between 6-12 more control experiments to complete this dataset, and verify that an abbreviated test paradigm captures the required data. We will present our results to the

Society for Neurosciences this fall (see abstract in next section). It is also imperative that we record as quickly as possible from the largest sample of post-centrifugation afferents. We have determined that incremental steps head angle orientation can be raised from 15° to 30°, thus reducing in half the recording time needed to obtain the afferent's directional selectivity and the maximum and minimum sensitivity.

Figures 1-3 show the battery of acceleration tests delivered to characterize the directional sensitivity of the afferent to inertial and gravitational accelerations. As evident in the figures the control utricular afferent is highly directionally tuned and its maximum and minimum sensitivity can be accurately calculated. These properties of the utricular afferent will play strongly into determining the structural and functional changes that occur following centrifugation. Figure 1 gives the responses of a utricular afferent to controlled translational accelerations. The afferent was recorded on the right side on the animal, and using the right-hand rule (lab convention) acceleration directed out the animal's snout are at 0°, out the contralateral labyrinth at 90°, out the tail at 180°, and out the ipsilateral labyrinth at 270°. Panel A provides the raw records. Upper trace is the animal's head orientation (yaw axis) at -90° (=270°), and the accelerations are thus delivered along the interaural axis. The next traces are the linear acceleration (in cm/s2), cycle sync ticks, the linear velocity (in cm/s), and the firing rate (in imp/s) of the afferent. The cycles of stimulation are averaged and presented in phase histograms of panel B. The upper histogram is the averaged firing rate of the afferent with the least-squares fit of the 1st harmonic component of the response (dotted curve). The lower histogram is the averaged acceleration stimulus (in g) with the least-squares fit of the 1st harmonic component of the stimulus (dashed curve). The stimulus was 2 Hz acceleration at ± 0.02g. The averaged response gain was 4658 imp/s/g and the peak of the response lead the peak of the acceleration at that angle by 43.4°. The directional sensitivity of the afferent is shown in panels C and D. In panel C the response gain (in imp/s/g) and response phase with respect to peak stimulus acceleration (in °) are plotted as a function of head angle (in °). The data points were obtained by applied the acceleration stimulus and rotating the fish about the Earth's vertical in 15° steps. A cosine function was applied (with equal weighting) to the gain and phase of the response to determine the degree of spatial tuning and the sensitivities: maximum sensitivity was 4803.5 imp/s/g at a head angle of 90° (and 270°) and the minimum was 0 imp/s/g at a head angle of 0° (and 180°). The Smin/Smax is equal to 0. A graphic representation of the data is given in panel D. The responses (in imp/s/g) with respect to the naso-occipital (NO) axis are plotted against the same responses with respect to the interaural (IA) axis. The green line with arrowheads shows the orientation of the best response - along the IA axis. (This is the most common response and agrees with the prevalent morphological polarization of the utricular hair cell bundle. Other directional sensitivities are also present.) The fact that the 2 red curve fits converge at zero indicated the tightness of the directional sensitivity (for example, the responses of an afferent to acceleration without directional selectivity would fall on a single, open function).

Figure 2 gives the responses of the same afferent to dynamic tilts and clearly shows the correspondence between the inertial and gravitational accelerations. The panels are similar in design to those presented in Fig.1. Following the right-hand rule and lab convention the fish's snout is straight forward at 0° and 180°; in this orientation the tilt is a pure roll stimulus about the fish's long axis. At 90° and 270° the tilts are pure pitch stimuli. Panel A provides the raw data: head orientation (yaw) at -30° or 330°, tilt velocity (in °/s), cycle sync ticks, tilt position (in °), and firing rate (in imp/s). The averaged response over 46 stimulus cycles is shown in panel B for the 2 Hz tilt at ±0.11° displacement: 445.4 imp/s/° and a phase lead of 49.3°. The directional selectivity of the afferent to dynamic tilt is given in panels C and D (same format as Fig. 1). Again, the afferent is highly tuned to this stimulus: maximum and minimum sensitivity are 793.9 imp/s/° at 19.7° (and 199.7°) and 45.6 imp/s/° at 109.7° (and 289.7°), respectively (Smin/Smax = 0.057), and the orientation is 19.7° (green line with arrowheads in panel D) off pure roll.

Figure 3 shows the responses of the same afferent in Figs. 1 and 2 to static displacements of the fish about the long axis (roll stimuli). For this afferent maximum responses were observed for translational accelerations directed out the contralateral labyrinth (Fig. 1), for dynamic roll tilts with ipsilateral side down (Fig. 2), and for static displacements moving toward ipsilateral displacement (Fig. 3). In Fig. 3 the upper trace is the displacement of the fish (ipsilateral or right side down is positive, contralateral or left side down is negative), and the bottom trace is the afferent firing rate over the 12-minute record. The insert shows the dynamic roll responses before the delivery of static displacement stimuli. The afferent gave an initial large excitatory response to ipsilateral displacement, followed by an adaptive decline in firing rate along two separate time courses - one fast with a dominant time constant of 4.3 sec and one slow with a dominant time constant of 44.6 sec. These data agree closely with recent results of hair cell receptor potential and afferent firing rate in the horizontal semicircular canal following constant copular displacement after canal lumen block (Rabbitt RD, Boyle R, Holstein GR, and Highstein SM (2005) Hair-cell versus afferent adaptation in the semicircular canals. J. Neurophysiol. 93: 424-36). The data on the control responses of utricular afferents will be presented at the Society for Neurosciences in November of this year, and the results will described in a separate publication. These data provide the benchmarks in response sensitivity and spatio-temporal properties required to accurately evaluate the utricular response to centrifugation.

2) Control Horizontal Semicircular Canal Afferent Characterization.

We have accepted a suggestion made by the study section review committee to include the study of the horizontal semicircular canal afferents. We found this to be a valuable and helpful suggestion. During centrifugation, the animal is subjected not only to a linear acceleration component due to the eccentricity of the animal with respect to the axis of rotation, but also to an angular acceleration stimulus. For this reason on-center control experiments are conducted to parse out the influence of the two forces. The question to be directly addressed is whether the semicircular canal undergoes structural and functional changes as a result of prolonged angular rotation. We have established the protocol that we will use to study the afferent responses, and have recorded to date the responses of 87 horizontal canal afferents. In addition we can now successfully record 25-35 canal afferents in two hours. As we previously described (Boyle R and Highstein SM (1990) Resting discharge and response dynamics of horizontal semicircular afferents of the toadfish, Opsanus tau. J. Neuroscience 10: 1557-1569), the canal afferents show a continuum of responses from velocity sensitivity to acceleration sensitivity. Our protocol uses frequencies from 0.2 Hz to 3 Hz, and will allow us to capture these responses. Results show a velocity-sensitive canal afferent and the lower panels show a more acceleration-sensitive canal afferent. Panels A1and B1 provide the raw data (acceleration, velocity and position of the stimulus, and the firing rate), A2 and B2 give the averaged responses in the form of a phase histogram and print the pertinent data for both the stimulus and the response, and A3 and B3 show the afferents' excitatory responses to electrical activation of the efferent vestibular system. Although we have already defined these responses, we needed to collect these data on our new system to permit direct correspondence with the centrifuged fish. On-center control experiments will also be performed. In the centrifugation experiments we will first study the utricular afferents (our principal goal) and then the canal afferents.

Task Progress:

3) Morphology of Utricular Afferents.

We are currently completing the morphological evaluation of the utricle at the scanning and transmission electron microscopic levels. This is a supplemental study that will benefit the analysis and interpretation of our current and projected results. This work will be written for publication in an anatomical journal.

We have intracellularly labeled utricular afferents in four separate experiments; a fifth labeling experiment is scheduled for May 25th. These data will serve as the control for the determining the regulation of the hair cell synaptic bodies and synaptic formation in fish subjected to centrifugation. In these experiments the afferent fiber is penetrated with a glass micropipette filled with a 3% solution of neurobiotin or biocytin. The responses of the afferent to short translational stimulations and electrical efferent stimulation are collected during intracellular penetration, and the label is iontophorezed into the fiber using positive current. Only one afferent is labeled in the fish to ensure positive identification. The utricle is then removed after several hours, fixed, and then immunohistochemically treated to visualize the label.

We are currently evaluating the labeled utricular afferents migrating through the supporting cell zone of the macula and into the hair cell zone at its base with thin sections (180 nm) sections at the transmission electron microscopic level. To develop a 3D reconstruction of the afferent a mosaic is first constructed at a reduced magnification to serve as a guide. The tissue is viewed at higher magnifications to identify the synaptic bodies in contact with the afferent processes. The mosaic technique is also used to obtain the density of synaptic bodies in the utricle, and this analysis is underway in control utricles. This effort is very laborious, and we are systematically assembling the TEM images for each afferent. The reconstructions will be completed in both the control and experimental fish.

Of the 12 hair cells in the utricular afferent, the synaptic bodies in all but one hair cell were exclusively associated with the labeled afferent. The dendritic process of the utricular afferent follows a line or beam along a set of hair cells. The hair cells share a similar morphological polarization of their hair bundles.

4) Modeling.

Significant effort has been made towards advancing our modeling of the utricular response to altered gravity. The major emphasis of our modeling effort will be directed towards interpreting the centrifugation results that will soon begin. Prof. Kondrachuk will attend the joint 9th Life Science Symposium of European Space Agency (ESA) and the 26th Annual Meeting of International Society of Gravitational Physiology (ISGP) and present “The role of possible feedback mechanisms in the effects of altered gravity on formation and function of gravireceptors of mollusks and fish.” The goals of this presentation are: 1) to review and compare the existing experimental data concerning the effects of altered gravity on the structure and function of gravireceptors in mollusks and fish; 2) to analyze possible mechanisms, structures and hierarchy of feedbacks that regulate formation and function of gravireceptors in mollusks and fish in normal and altered gravity conditions; 3) to discuss the reasons for difference between the feedbacks in the developing and mature gravireceptors and the possible role of the efferent system in their self-regulation; 4) to propose qualitative model(s) of the effects of altered gravity on gravireceptors of mollusks and fish in the context of the feedback problem; and 5) to formulate possible experiments that may be useful to test the model proposed. The presentation will be written in a 3-page conference proceeding. 5) Centrifugation Experiments.

We have been working continuously to get the experimental hardware design complete and begin the centrifugation cycles since before the start of the project. As originally planned, we would use the 20-foot centrifuge at Ames. This was a challenging effort as functional requirement to interface with the facilities had to be made from new or we would have to develop a complete standalone unit including the cooling system and power. We choose to create a standalone unit.

Six months ago it was stated that we could shift the project to the Ames ISS 8-foot centrifuge. This change was made to help facilitate the project because the 8-foot centrifuge a) has a cold liquid loop that could serve to regulate the aquarium water temperature and b) has only minimal user demands thus allowing the project to use the centrifuge without schedule restriction. However, more effort was needed for the facility utilization office to test both the centrifuge and its ISS specific cooling loop and for us to change the aquarium configuration to accommodate a heat exchanger (instead of a standalone system) and reduce allowable weight.

We have managed and conducted weekly status meetings with the facility engineers and experiment team to make the system integration a workable configuration. We also brought on a mechanical engineering graduate student to help in the design and test of the aquarium components, and particularly the maintenance of water temperature at 13-14°C and configuration design. The water temperature control for the fish is very stable and we are pleased with the final thermal results. This was the largest technical effort because of the power and weight constraints. Safety and structural stability tests have been completed on the aquarium itself. The assembly of the final design experiment hardware system started on 5/17/2005. The first run (without fish) is scheduled for early June. This test will conclude the final structure and safety tests required. Once this final hardware shakedown is completed, we will begin the first science experimental run with fish in July 2005.

New Technology -- NONE

Presentations 2005

1) Society for Neuroscience Nov. 12-16, 2005; Washington, DC

Abstract accepted for presentation:

“Response properties of utricular nerve afferents to gravitational and translational accelerations in the toadfish.”

Y. Popova, A. Twombly, J. Varelas, R. Ehsanian, A. Kondrachuk and R. Boyle.

Ames Research Center, NASA, Moffett Field, CA 94035 USA.

The utricle is an otolith organ in the vertebrate inner ear that senses gravity and inertial acceleration of the head. The detection of these forces relies on the otolith mass being weighted in Earth’s 1g. Under weightless conditions experienced in space flight the sensation is altered, and structural and functional changes are known to occur in the utricle. These adaptive changes pose a serious challenge to adequate sensorimotor function during transitions from one gravity level to another. The aim of the present study was to characterize the firing rate response of the utricular afferents to controlled gravitoinertial accelerations in the anesthetized oyster toadfish, Opsanus tau. The data will provide

the benchmark for interpretation of subsequent results in altered gravity experiments, such as centrifugation, modeling the adaptive changes and efficacy of countermeasures. Translational accelerations along the earth's horizon were applied as the animal was rotated about the earth's vertical at 15° intervals over 360°. Rotations about the earth's horizon were delivered in a similar fashion to apply pitch to roll stimuli; static reorientations in gravitational field were also applied. Single fiber recording techniques were used to acquire the firing rate responses of the utricular afferents. Responses to both translational and gravitational accelerations are highly directionally selective. This agrees closely with the afferent's tight dendritic field in the sensory epithelium. Firing rate recovery following a ramp displacement of the animal at an orientation that elicits a maximum or minimum sensitivity had a dominant time constant in the range of 30-50 sec. Afferents were also studied during combined electric shock activation of the efferent vestibular pathway. The firing rate was facilitated (most common), unresponsive or disfacilitated; when responsive to efferent stimulation, the afferent sensitivity to an applied acceleration was reduced. Support Contributed By: NIDCD DC01837 and NASA 03-OBPR-04

2) The joint 9th Life Science Symposium of European Space Agency (ESA) and the 26th Annual Meeting of International Society of Gravitational Physiology (ISGP) "Life in Space for Life on Earth" 26 June-1 July 2005; Cologne, Germany.

The work is currently being written for a 3-page conference publication. Paper will be submitted in July 2005. The abstract accepted for presentation is:

"The role of possible feedback mechanisms in the effects of altered gravity on formation and function of gravireceptors of mollusks and fish."

Alexander V. Kondrachuk 1 and Richard D. Boyle 2 1 Dept. Theoretical Physics, Institute of Physics, Natl. Acad. Sci., Kiev, Ukraine, kondr@kondr.kiev.ua 2 BioVIS Technology Center and Vestibular Research Facility, Ames Research Center, NASA, Moffett Field, CA 94035-1000, USA. richard.boyle@nasa.gov

Altered gravity is known to influence the state of gravireceptors. Although these effects are more pronounced during the earlier stages of organism development, structural changes also take place in mature gravireceptors. Despite the number of experiments, the results have not led to a clear interpretation. Many interpretations were guided by popular hypothesis that says that regulation of the growth of otoliths or mass of statoconia, i.e., the test mass upon which gravity and linear acceleration act, is based on the weight of the mass. Thus, a larger-than-normal mass should be produced in a reduced-g environment and a smaller-than-normal mass would be produced in hyper-g. This hypothesis is based on the idea of feedback (or feedbacks) and seems to be intuitively clear; however, its grounds and subsequences were not sequentially analyzed.

In the past years there was significant progress in discovering the details of the early stages of development and function of gravireceptors in mollusks and fish, particularly in the conditions of hyper- and microgravity. From an evolutionary point of view we may expect that the main processes of the earlier stages of formation of gravireceptors in mollusks and fish are consistent with the development of gravireceptors in higher animals. An important consideration is that the mechanisms of self-regulations might differ between the developing and the mature organisms, and thus their respective sensitivities to transitions of gravity are different. If this is indeed the case, then different structural and functional changes in their gravireceptors will result by alteration of gravity. The goals of the present work are: 1) to review and compare the existing experimental data concerning the effects of altered gravity on the structure and function of gravireceptors in mollusks and fish; 2) to analyze possible mechanisms, structures and hierarchy of feedbacks that regulate formation and function of gravireceptors in mollusks and fish in normal and altered gravity conditions; 3) to discuss the reasons for difference between the feedbacks in the developing and mature gravireceptors and the possible role of the efferent system in their self-regulation; 4) to propose qualitative model(s) of the effects of altered gravity on gravireceptors of mollusks and fish in the context of the feedback problem; and 5) to formulate possible experiments that may be useful to test the model proposed. Support Contributed By: NASA 03-OBPR-04

3) For Publication in Hearing Research (preliminary abstract). Prof. Kondrachuk and I have exchanged numerous edits to the manuscript; we're at revision 12. We will first finish the conference paper (item 2 above) for the July submission date, and then complete this paper. The manuscript currently has 19 text pages and 2 figures.

"On the mechanisms of stimulation of the hair cell bundles in otolithic organs"

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Abstract

The interaction of the hair cell bundle (HCB) with the otolithic membrane (OM) caused by linear acceleration or gravity initiates the transduction cascade and the generation of the HCB receptor potential that can trigger a neural afferent discharge. Therefore, the response of the OM components that surround the HCBs to the applied acceleration is important for the transfer of mechanical input to the HCB level. Experimental investigations of the function of hair cell bundles are based on artificial mechanical stimulation of the HCB by probes and fluid-jets. The analysis of hair cells function must consider the movements of the HCB and its surroundings as a single system. Meanwhile, a detailed picture of the nature of the interactions between the HCBs and the surrounding gel and nearby structures remains unclear. The present work considers the question: whether the conditions of artificial stimulation justify the conditions of natural stimulation that follow from the morphological data on the structure of otolithic membrane and results of modeling The HCBs can be divided into two general groups based on their type of contact with environment: 1) those with one or several point contacts between the highest cilia and the otoconial layer, as may occur in the frog saccule, and 2) those with spatially continuous contacts with the environment. Freestanding HCBs are the example of the second group. The purpose of the present work is to estimate the parameters of the artificial stimulations and to analyze their correspondence to natural stimulation of the HCB in otoliths. This analysis is based on results of the modeling of transformations of mechanical input in the following sequence: acceleration of the otolithic membrane, displacement of the OM gel layer, deflection of hair cell bundle, deformation of the ciliary tip-links, and formation of a temporal pattern of polarization in the framework of simplified models of the HCB+OM interaction corresponding to point and spatially continuous contacts of the HCBs with the OM. The comparison of experimental and modeling data indicates that probe stimulation cannot imitate the effects of spatially continuous contacts, but may imitate the natural stimulation of the HCBs with point contacts. Stimulation of the HCB by fluid jet mimics the fluid-like gel interaction with the HCB, but

application of the fluid jet is restricted by the low viscosity of the solution. The parameters of fluid jet stimulation indicate that inertial force, rather than viscous force, is responsible for the HCB deflection in these experiments. This could be verified by direct measurements of the parameters of fluid-jet stimulation. The possible experiments that may improve the adequacy of artificial stimulation of the HCB to natural one are discussed. Support Contributed By: NASA 03-OBPR-04

4) The P.I. was the Keynote Speaker, California State University at Sacramento Conference on Pre-medicine Education and Training, Sacramento, CA; Feb. 2005. This was an oral PowerPoint presentation of Space Exploration and the role of the medical and biomedical sciences in advancing crew health. The nervous system adaptation to the space environment, and particularly the structural and functional changes occurring in the vestibular labyrinth and their influence on sensorimotor coordination, was a principal theme.

Future Research Plan.

We are following closely the original research plan. Additional control experiments are planned, but these will constitute a minor percent effort. The priority is to conduct the centrifugation experiments, and answer the prime question: can we accurately and precisely quantify the influence of exposure to hyper-g on the utricular end organ. Based on interpretation of data from human, rat and toadfish, we expect to find reduced response sensitivity to gravitoinertial accelerations in the utricular afferents and a reduction in the number of synaptic bodies in the hair cells. The experimental plan was designed to extract the influence of both the time course and g-level on the utricular response. First experimental runs will start in July. The 8-foot centrifuge has 4 boxes at a fixed radius from the midline. Consequently, more experimental runs will be needed; however, our project should have unlimited access to the centrifuge. The first run will begin in July 2005. We are scheduling a 3-g run of 25-day duration to obtain data in the following groupings: 4-5 day (4 fish), 8-9 days (3 fish) and 16-17 days (3 fish). Subsequent experimental runs will extend the results in each grouping, including 1 and 2 day groups, and their counterparts at 1.5 g exposure. In these experiments we will first focus on obtaining adequate recordings from utricular nerve afferents, followed by collect data from the horizontal semicircular canal afferents.

In addition to the electrophysiology experiments we follow our plan to determine the density of synaptic bodies in hair cells in control and experimental animals. This work is under way. I am currently searching for a student to help in this effort. Intracellular labeling will be performed on a selected utricular afferent in each experimental fish for the anatomical study.

The modeling effort led by Alexander Kondrachuk will proceed as planned. Prof. Kondrachuk will return for a 2-month stay in October. During this period, we will work closely on evaluating the experimental data and developing the self-learning and feedback models. Prof. Kondrachuk is also an expert on utricular anatomy, particularly the otolith membrane and mass. With his input we will extend our morphological study of the utricle to include a more detailed description of the otolith membrane and its association with the hair bundles, the otolith mass, and the surrounding epithelium.

We are also starting the electrical and mechanical construction of the on-center control device. The 8-foot centrifuge has an inadequate on-center control structure for our purposes: the available size of the area is too limited and the location places a significant g-load on the fish. Thus, we are constructing our own system based on existing hardware. This is not a difficult task, but it will take time to construct. Importantly, the device will permit the placement of the animal truly on-center for the rotation controls.

Bibliography Type:	Description: (Last Updated: 09/17/2021)
Abstracts for Journals and Proceedings	Popova, Y.;Twombly, A.;Varelas, J.; Ehsanian, R.; Kondrachuk, A.; Boyle, R. "Response properties of utricular nerve afferents to gravitational and translational accelerations in the toadfish." Society for Neuroscience Nov. 12-16, 2005; Washington, DC TBD , Nov-2005
Abstracts for Journals and Proceedings	Kondrachuk, A.V.; Boyle, RD "The role of possible feedback mechanisms in the effects of altered gravity on formation and function of gravireceptors of mollusks and fish" The joint 9th Life Science Symposium of European Space Agency (ESA) and the 26th Annual Meeting of International Society of Gravitational Physiology (ISGP) “Life in Space for Life on Earth” 26 June-1 July 2005; Cologne, Germany. TBD , Aug-2005